TM Forum Specification

Resource Usage Management User Guide

TMF771

Maturity Level: General Availability (GA)	**Team Approved Date: 05-Jul-2024**
Release Status: Stable	**Approval Status: Team Approved**
Version 5.0.0	**IPR Mode: RAND**

© TM Forum 2024. All Rights Reserved.

NOTICE

Copyright © TM Forum 2024. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the TM FORUM IPR Policy, must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an "AS IS" basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

Direct inquiries to the TM Forum office:

181 New Road, Suite 304
Parsippany, NJ 07054, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

Table of Contents

NOTICE . . . i
Introduction . . . 1
Sample Use Cases . . . 2
Support of polymorphism and extension patterns . . . 3
RESOURCE MODEL . . . 4
Managed Entity and Task Resource Models . . . 4
ResourceUsage resource . . . 4
ResourceUsageSpecification resource . . . 11
Notification Resource Models . . . 19
ResourceUsage . . . 20
ResourceUsageSpecification . . . 24
API OPERATIONS . . . 29
Operations on ResourceUsage . . . 29
List or find ResourceUsage objects . . . 29
Retrieves a ResourceUsage by ID . . . 31
Creates a ResourceUsage . . . 32
Updates partially a ResourceUsage . . . 35
Deletes a ResourceUsage . . . 40
Operations on ResourceUsageSpecification . . . 40
Retrieves a ResourceUsageSpecification by ID . . . 40
List or find ResourceUsageSpecification objects . . . 41
Creates a ResourceUsageSpecification . . . 42
Updates partially a ResourceUsageSpecification . . . 44
Deletes a ResourceUsageSpecification . . . 48
API NOTIFICATIONS . . . 49
Register listener . . . 49
Unregister listener . . . 50
Publish Event to listener . . . 50
Acknowledgements . . . 52
Release History . . . 52
Contributors to Document . . . 52

© TM Forum 2024. All Rights Reserved.

Introduction

This document is the user-guide of the REST API for Resource Usage Management. It includes the data model definition as well as all available operations for the Usage and UsageSpecification resources.

The Resource Usage Management API provides a standardized mechanism for resource usage management such as the creation, update, retrieval, import and export of a collection of usages. A resource usage is an occurrence of employing a Resource, Resource Usage can influence of Service Usage and/or Product Usage.

The Resource Usage API manages measurement usages. For example, it allows a service provider to count usage of Antenna for 5G slice:

- To retrieve usage
- Collect Usage and update usage of Service and Product

The Resource Usage API manages the Resource Usage resource:

- A Resource Usage represents an event of usage. It is comprised of characteristics which define all the attributes known for that particular type of usage.
- The main attributes are its id, usageDate, usageType, it's usageType related UsageSpecification, characteristics (name/value pairs), status, related Parties
- A Resource Usage follows a simple lifecycle which is presented later for illustrative purposes, defining such aspects as whether it is rated, rerated or billed.

The Resource Usage API performs the following operations on the Usage resource:

- Retrieval of a Resource Usage or a collection of Resource Usage, depending on filter criteria
- Partial update of a Resource Usage or of a collection of Resource Usage
- Creation of a Resource usage

© TM Forum 2024. All Rights Reserved.

Sample Use Cases

A few simple use-cases are presented:

- Creation of a simple ResourceUsageSpecification, defining the characteristics a data usage in Antenna for 5G slice; listing ResourceUsageSpecification's at version 2.5; retrieving a single ResourceUsageSpecification based on it's id and ultimately its deletion based on the same id.
- Creation of a simple ResourceUsage relating to a voice call retrieval; listing Resource Usage of usageType "DATA"; retrieving a specific Resource Usage based on it's id; updating a Resource Usage and ultimately deleting this Usage based on the same id.

© TM Forum 2024. All Rights Reserved.

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

© TM Forum 2024. All Rights Reserved.

RESOURCE MODEL

Managed Entity and Task Resource Models

ResourceUsage resource

An occurrence of usage on a Resource,derived from various Service usages. It is comprised of characteristics, which represent attributes of service usage.

Resource model

© TM Forum 2024. All Rights Reserved.

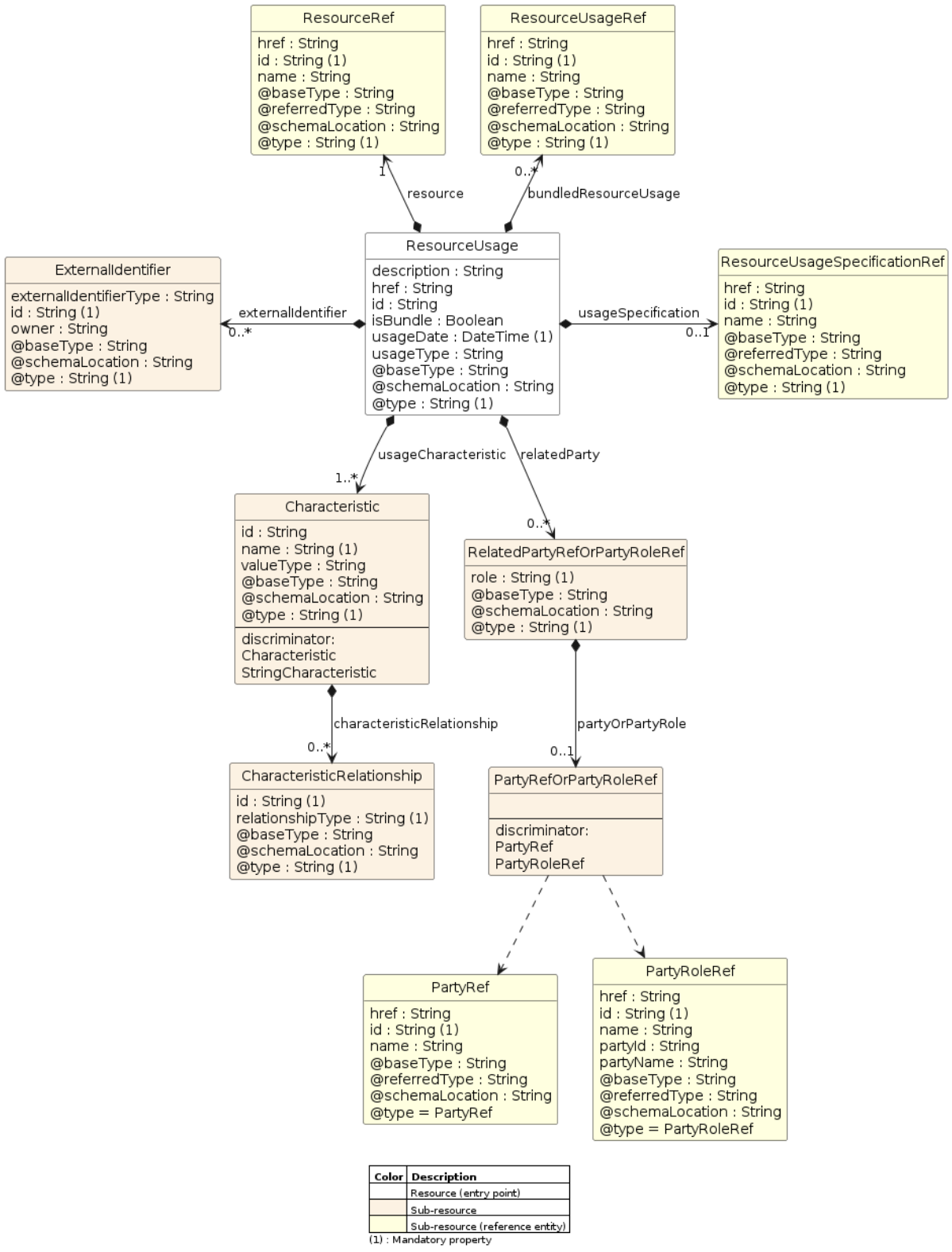


Figure 1 - ResourceUsage

Field descriptions

ResourceUsage fields

© TM Forum 2024. All Rights Reserved.

bundledResourceUsage	A ResourceUsageRef. ResourceUsage reference. ResourceUsage is usage event for Resource.
description	A String. Description of resource usage.
externalIdentifier	An ExternalIdentifier. An identification of an entity that is owned by or originates in a software system different from the current system, for example a ProductOrder handed off from a commerce platform into an order handling system. The structure identifies the system itself, the nature of the entity within the system (e.g. class name) and the unique ID of the entity within the system. It is anticipated that multiple external IDs can be held for a single entity, e.g. if the entity passed through multiple systems on the way to the current system. In this case the consumer is expected to sequence the IDs in the array in reverse order of provenance, i.e. most recent system first in the list.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
isBundle	A Boolean. IsBundle determines whether usage represents a single usage (false), or a bundle of usages specifications (true).
relatedParty	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
resource	A ResourceRef. Resource reference, for when Resource is used by other entities.
usageCharacteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic… Characteristic can be instanciated as * StringCharacteristic
usageDate	A DateTime. Date of usage.
usageSpecification	A ResourceUsageSpecificationRef. ResourceUsageSpecification reference. ResourceUsageSpecification is a detailed description of a usage event that are of interest to the business and can have charges applied to it. It is comprised of characteristics, which define all attributes known for a particular type of usage.
usageType	A String. Type of usage.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

© TM Forum 2024. All Rights Reserved.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceUsageSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instanciated as * StringCharacteristic
value	This property is present in subclasses

StringCharacteristic sub-resource fields

© TM Forum 2024. All Rights Reserved.

characteristicRelationship	A CharacteristicRelationship. Another Characteristic that is related to the current Characteristic;.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceUsageRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

© TM Forum 2024. All Rights Reserved.

@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.

© TM Forum 2024. All Rights Reserved.

@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ResourceUsage' resource object.


```
{
    "@type": "ResourceUsage",
    "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
    "id": "5130d5131a23",
    "usageDate": "2024-08-25T12:19:14Z",
    "usageType": "VOICE",
    "usageCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "originatingCountryCode",
            "valueType": "string",
            "value": "43"
        },
        {
            "@type": "StringCharacteristic",
            "name": "destinationCountryCode",
            "valueType": "string",
            "value": "49"
        },
        {
            "@type": "StringCharacteristic",
            "name": "destinationNumber",
            "valueType": "string",
            "value": "170123456789"
        },
        {
            "@type": "StringCharacteristic",
            "name": "duration",
            "valueType": "string",
            "value": "20"
        },
        {
            "@type": "StringCharacteristic",
            "name": "startDateTime",
            "valueType": "datetime",
            "value": "2024-08-25T12:19:14Z"
        },
        {
            "@type": "StringCharacteristic",
            "name": "endDateTime",
            "valueType": "datetime",
            "value": "2024-08-25T12:19:34Z"
        }
    ],
    "relatedParty": [
```


© TM Forum 2024. All Rights Reserved.

```
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Individual",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "http://api.csp.com/tmf-api/partyManagement/individual/8a41-
d6451fe98963",
                "id": "8a41-d6451fe98963"
            }
        },
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Organization",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "http://api.csp.com/tmf-
api/partyManagement/organization/a396-5c90a41a23ea",
                "id": "a396-5c90a41a23ea"
            }
        }
    ],
    "resource": {
        "@type": "ResourceRef",
        "href": "https://api.csp.com/tmf-api//resource/qerr-475",
        "id": "qerr-475",
        "name": "Voice resource"
    },
    "usageSpecification": {
        "@type": "ResourceUsageSpecificationRef",
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
        "id": "af59-b504c742900e",
        "name": "VoiceCall"
    }
}
```


ResourceUsageSpecification resource

A detailed description of a Resource usage event that are of interest to the business and can have charges applied to it. It is comprised of characteristics, which define all attributes known for a particular type of usage.

Resource model

© TM Forum 2024. All Rights Reserved.

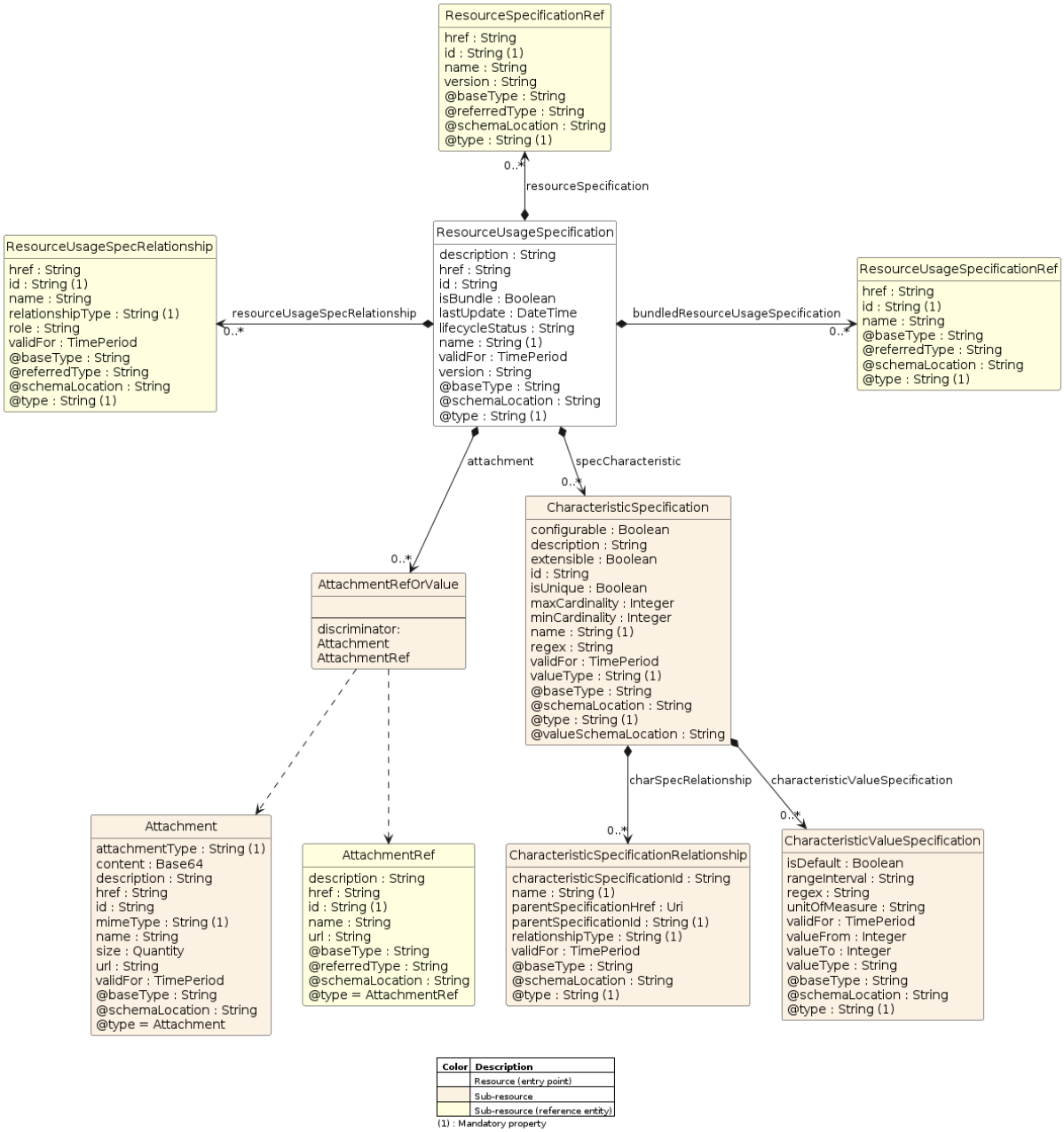


Figure 2 - ResourceUsageSpecification

Field descriptions

ResourceUsageSpecification fields

attachment	An AttachmentRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Attachment entity and not the AttachmentRefOrValue class itself.
bundledResourceUsageSpecification	A ResourceUsageSpecificationRef. ResourceUsageSpecification reference. ResourceUsageSpecification is a detailed description of a usage event that are of interest to the business and can have charges applied to it. It is comprised of characteristics, which define all attributes known for a particular type of usage.

© TM Forum 2024. All Rights Reserved.

description	A String. Description of the specification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
isBundle	A Boolean. IsBundle determines whether specification represents a single specification (false), or a bundle of specifications (true).
lastUpdate	A DateTime. Date and time of the last update of the specification.
lifecycleStatus	A String. Used to indicate the current lifecycle status of this catalog item.
name	A String. Name given to the specification.
resourceSpecification	A ResourceSpecificationRef. Resources are physical or non-physical components (or some combination of these) within an enterprise's infrastructure or inventory. They are typically consumed or used by services (for example a physical port assigned to a service) or contribute to the realization of a Product (for example, a SIM card). They can be drawn from the Application, Computing and Network domains, and include, for example, Network Elements, software, IT systems, content and information, and technology components. A ResourceSpecification is an abstract base class for representing a generic means for implementing a particular type of Resource. In essence, a ResourceSpecification defines the common attributes and relationships of a set of related Resources, while Resource defines a specific instance that is based on a particular ResourceSpecification.
resourceUsageSpecRelationship	A ResourceUsageSpecRelationship. A dependency, substitution or exclusivity relationship between/among resource usage specifications.
specCharacteristic	A CharacteristicSpecification. This class defines a characteristic specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceUsageSpecRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

© TM Forum 2024. All Rights Reserved.

relationshipType	A String. Type of relationship such as dependency, substitution or exclusivity.
role	A String. The association role for this service specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceUsageSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

© TM Forum 2024. All Rights Reserved.

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicSpecification sub-resource fields

© TM Forum 2024. All Rights Reserved.

charSpecRelationship	A CharacteristicSpecificationRelationship. An aggregation, migration, substitution, dependency or exclusivity relationship between/among Characteristic specifications. The specification characteristic is embedded within the specification whose ID and href are in this entity, and identified by its ID.
characteristicValueSpecification	A CharacteristicValueSpecification. Specification of a value (number or text or an object) that can be assigned to a Characteristic. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : BooleanArrayCharacteristicValueSpecification, BooleanCharacteristicValueSpecification, FloatArrayCharacteristicValueSpecification, FloatCharacteristicValueSpecification, IntegerArrayCharacteristicValueSpecification, IntegerCharacteristicValueSpecification, MapArrayCharacteristicValueSpecification, MapCharacteristicValueSpecification, NumberArrayCharacteristicValueSpecification, NumberCharacteristicValueSpecification, ObjectArrayCharacteristicValueSpecification, ObjectCharacteristicValueSpecification, StringArrayCharacteristicValueSpecification, StringCharacteristicValueSpecification…
configurable	A Boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A String. A narrative that explains the CharacteristicSpecification.
extensible	A Boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A String. Unique ID for the characteristic.
isUnique	A Boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An Integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An Integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A String. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A String. A rule or principle represented in regular expression used to derive the value of a characteristic value.

© TM Forum 2024. All Rights Reserved.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueType	A String. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
@valueSchemaLocation	A String. This (optional) field provides a link to the schema describing the value type.

CharacteristicSpecificationRelationship sub-resource fields

characteristicSpecificationId	A String. Unique identifier of the characteristic within the specification.
name	A String. Name of the target characteristic within the specification.
parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A String. Unique identifier of the parent specification containing the target characteristic.
relationshipType	A String. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.

© TM Forum 2024. All Rights Reserved.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ResourceUsageSpecification' resource object.


```
{
    "@type": "ResourceUsageSpecification",
    "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/9aa3-3e7737dd5080",
    "id": "9aa3-3e7737dd5080",
    "name": "Voice-Call-Specification",
    "description": "Specification for voice calls usage",
    "version": "2.5",
    "specCharacteristic": [
        {
            "@type": "CharacteristicSpecification",
            "name": "originatingCountryCode",
            "valueType": "string",
            "description": "country code of the caller",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "originatingNumber",
            "valueType": "string",
            "description": "phone number of the caller",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "destinationCountryCode",
            "valueType": "string",
            "description": "country code of the called party",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        },
        {
```


© TM Forum 2024. All Rights Reserved.

```
            "@type": "CharacteristicSpecification",
            "name": "destinationNumber",
            "valueType": "string",
            "description": "phone number of the called party",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "duration",
            "valueType": "string",
            "description": "duration of the call",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "startDateTime",
            "valueType": "dateTime",
            "description": "startdate and starttime of the call",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "endDateTime",
            "valueType": "dateTime",
            "description": "enddate and endtime of the call",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        }
    ],
    "validFor": {
        "startDateTime": "2022-02-20T17:00:00Z",
        "endDateTime": "2099-01-11T00:00:00Z"
    }
}
```


Notification Resource Models

6 notifications are defined for this API.

Notifications related to ResourceUsage:

- Create Event
- Delete Event
- Attribute Value Change Event

Notifications related to ResourceUsageSpecification:

- Create Event
- Delete Event

© TM Forum 2024. All Rights Reserved.

- Attribute Value Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

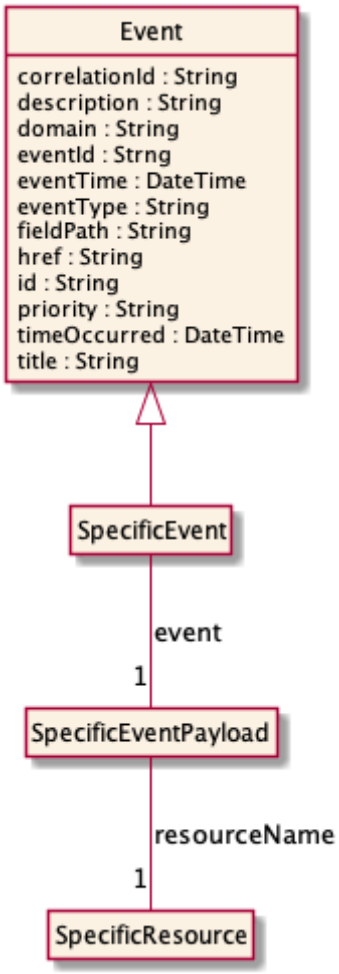


Figure 3 Notification Pattern

ResourceUsage

Create Event

Message example for ResourceUsageCreateEvent event

```
Content-Type: application/json

{
    "@type": "ResourceUsageCreateEvent",
    "@baseType": "Event",
    "correlationId": "fe9b29c9-a31c",
```

© TM Forum 2024. All Rights Reserved.

```
    "domain": "Commercial",
    "title": "ResourceUsageCreateEvent",
    "description": "ResourceUsageCreateEvent illustration",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:19:14Z",
    "source": {
        "@type": "ReportingResource",
        "id": "29",
        "name": "APP-171",
        "@referredType": "LogicalResource"
    },
    "reportingSystem": {
        "@type": "ReportingResource",
        "id": "182",
        "name": "APP-116",
        "@referredType": "LogicalResource"
    },
    "eventId": "4b9d-b37c-204557d70b97",
    "eventTime": "2024-08-25T12:19:14Z",
    "eventType": "ResourceUsageCreateEvent",
    "event": {
        "usage": {
            "@type": "ResourceUsage",
            "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
            "id": "5130d5131a23",
            "usageDate": "2024-08-25T12:19:14Z",
            "usageType": "VOICE",
            "usageCharacteristic": [
                {
                    "@type": "StringCharacteristic",
                    "name": "originatingCountryCode",
                    "valueType": "string",
                    "value": "43"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "destinationCountryCode",
                    "valueType": "string",
                    "value": "49"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "destinationNumber",
                    "valueType": "string",
                    "value": "170123456789"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "duration",
                    "valueType": "string",
                    "value": "20"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "startDateTime",
                    "valueType": "datetime",
                    "value": "2024-08-25T12:19:14Z"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "endDateTime",
                    "valueType": "datetime",
```


© TM Forum 2024. All Rights Reserved.

```
                    "value": "2024-08-25T12:19:34Z"
                }
            ],
            "relatedParty": [
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "Individual",
                    "partyOrPartyRole": {
                        "@type": "PartyRef",
                        "href": "http://api.csp.com/tmf-
api/partyManagement/individual/8a41-d6451fe98963",
                        "id": "8a41-d6451fe98963"
                    }
                },
                {
                    "@type": "RelatedPartyRefOrPartyRoleRef",
                    "role": "Organization",
                    "partyOrPartyRole": {
                        "@type": "PartyRef",
                        "href": "http://api.csp.com/tmf-
api/partyManagement/organization/a396-5c90a41a23ea",
                        "id": "a396-5c90a41a23ea"
                    }
                }
            ],
            "resource": {
                "@type": "ResourceRef",
                "href": "https://api.csp.com/tmf-api//resource/qerr-475",
                "id": "qerr-475",
                "name": "Voice resource"
            },
            "usageSpecification": {
                "@type": "ResourceUsageSpecificationRef",
                "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
                "id": "af59-b504c742900e",
                "name": "VoiceCall"
            }
        }
    }
}
```


Delete Event

Message example for ResourceUsageDeleteEvent event


```
Content-Type: application/json

{
    "@type": "ResourceUsageDeleteEvent",
    "@baseType": "Event",
    "correlationId": "fe9b29c9-a31c",
    "domain": "Commercial",
    "title": "ResourceUsageDeleteEvent",
    "description": "ResourceUsageDeleteEvent illustration",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:19:14Z",
    "source": {
        "@type": "ReportingResource",
        "id": "29",
```


© TM Forum 2024. All Rights Reserved.

```
        "name": "APP-171",
        "@referredType": "LogicalResource"
    },
    "reportingSystem": {
        "@type": "ReportingResource",
        "id": "182",
        "name": "APP-116",
        "@referredType": "LogicalResource"
    },
    "eventId": "4b9d-b37c-204557d70b97",
    "eventTime": "2024-08-25T12:19:14Z",
    "eventType": "ResourceUsageDeleteEvent",
    "event": {
        "usage": {
            "@type": "ResourceUsage",
            "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
            "id": "5130d5131a23"
        }
    }
}
```


Attribute Value Change Event

Message example for ResourceUsageAttributeValueChangeEvent event


```
Content-Type: application/json

{
    "@type": "ResourceUsageAttributeValueChangeEvent",
    "@baseType": "Event",
    "correlationId": "fe9b29c9-a31c",
    "domain": "Commercial",
    "title": "ResourceUsageAttributeValueChangeEvent",
    "description": "ResourceUsageAttributeValueChangeEvent illustration",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:19:14Z",
    "source": {
        "@type": "ReportingResource",
        "id": "29",
        "name": "APP-171",
        "@referredType": "LogicalResource"
    },
    "reportingSystem": {
        "@type": "ReportingResource",
        "id": "182",
        "name": "APP-116",
        "@referredType": "LogicalResource"
    },
    "eventId": "4b9d-b37c-204557d70b97",
    "eventTime": "2024-08-25T12:19:14Z",
    "eventType": "ResourceUsageAttributeValueChangeEvent",
    "event": {
        "usage": {
            "@type": "ResourceUsage",
            "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
            "id": "5130d5131a23",
            "usageCharacteristic": [
                {
```


© TM Forum 2024. All Rights Reserved.

```
                    "@type": "StringCharacteristic",
                    "name": "originatingCountryCode",
                    "valueType": "string",
                    "value": "43"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "destinationCountryCode",
                    "valueType": "string",
                    "value": "49"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "destinationNumber",
                    "valueType": "string",
                    "value": "170123456789"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "duration",
                    "valueType": "string",
                    "value": "20"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "startDateTime",
                    "valueType": "datetime",
                    "value": "2024-08-25T12:19:14Z"
                },
                {
                    "@type": "StringCharacteristic",
                    "name": "endDateTime",
                    "valueType": "datetime",
                    "value": "2024-08-25T12:19:34Z"
                }
            ]
        }
    }
}
```

ResourceUsageSpecification

Create Event

Message example for ResourceUsageSpecificationCreateEvent event

```
Content-Type: application/json

{
    "@type": "ResourceUsageSpecificationCreateEvent",
    "@baseType": "Event",
    "correlationId": "fe9b29c9-a31c",
    "domain": "Commercial",
    "title": "ResourceUsageSpecificationCreateEvent",
    "description": "ResourceUsageSpecificationCreateEvent illustration",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:19:14Z",
    "source": {
        "@type": "ReportingResource",
```

© TM Forum 2024. All Rights Reserved.

```
        "id": "29",
        "name": "APP-171",
        "@referredType": "LogicalResource"
    },
    "reportingSystem": {
        "@type": "ReportingResource",
        "id": "182",
        "name": "APP-116",
        "@referredType": "LogicalResource"
    },
    "eventId": "4b9d-b37c-204557d70b97",
    "eventTime": "2024-08-25T12:19:14Z",
    "eventType": "ResourceUsageSpecificationCreateEvent",
    "event": {
        "usageSpecification": {
            "@type": "ResourceUsageSpecification",
            "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/9aa3-3e7737dd5080",
            "id": "9aa3-3e7737dd5080",
            "name": "Voice-Call-Specification",
            "description": "Specification for voice calls usage",
            "version": "2.5",
            "specCharacteristic": [
                {
                    "@type": "CharacteristicSpecification",
                    "name": "originatingCountryCode",
                    "valueType": "string",
                    "description": "country code of the caller",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "originatingNumber",
                    "valueType": "string",
                    "description": "phone number of the caller",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "destinationCountryCode",
                    "valueType": "string",
                    "description": "country code of the called party",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "destinationNumber",
                    "valueType": "string",
                    "description": "phone number of the called party",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "duration",
                    "valueType": "string",
```


© TM Forum 2024. All Rights Reserved.

```
                    "description": "duration of the call",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "startDateTime",
                    "valueType": "dateTime",
                    "description": "startdate and starttime of the call",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "endDateTime",
                    "valueType": "dateTime",
                    "description": "enddate and endtime of the call",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                }
            ],
            "validFor": {
                "startDateTime": "2022-02-20T17:00:00Z",
                "endDateTime": "2099-01-11T00:00:00Z"
            }
        }
    }
}
```

Delete Event

Message example for ResourceUsageSpecificationDeleteEvent event

```
Content-Type: application/json

{
    "@type": "ResourceUsageSpecificationDeleteEvent",
    "@baseType": "Event",
    "correlationId": "fe9b29c9-a31c",
    "domain": "Commercial",
    "title": "ResourceUsageSpecificationDeleteEvent",
    "description": "ResourceUsageSpecificationDeleteEvent illustration",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:19:14Z",
    "source": {
        "@type": "ReportingResource",
        "id": "29",
        "name": "APP-171",
        "@referredType": "LogicalResource"
    },
    "reportingSystem": {
        "@type": "ReportingResource",
        "id": "182",
        "name": "APP-116",
        "@referredType": "LogicalResource"
    },
    "eventId": "4b9d-b37c-204557d70b97",
```

© TM Forum 2024. All Rights Reserved.

```
    "eventTime": "2024-08-25T12:19:14Z",
    "eventType": "ResourceUsageSpecificationDeleteEvent",
    "event": {
        "usageSpecification": {
            "@type": "ResourceUsageSpecification",
            "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/9aa3-3e7737dd5080",
            "id": "9aa3-3e7737dd5080",
            "name": "Voice-Call-Specification"
        }
    }
}
```


Attribute Value Change Event

Message example for ResourceUsageSpecificationAttributeValueChangeEvent event


```
Content-Type: application/json

{
    "@type": "ResourceUsageSpecificationAttributeValueChangeEvent",
    "@baseType": "Event",
    "correlationId": "fe9b29c9-a31c",
    "domain": "Commercial",
    "title": "ResourceUsageSpecificationAttributeValueChangeEvent",
    "description": "ResourceUsageSpecificationAttributeValueChangeEvent
illustration",
    "priority": "5",
    "timeOccurred": "2022-08-25T12:19:14Z",
    "source": {
        "@type": "ReportingResource",
        "id": "29",
        "name": "APP-171",
        "@referredType": "LogicalResource"
    },
    "reportingSystem": {
        "@type": "ReportingResource",
        "id": "182",
        "name": "APP-116",
        "@referredType": "LogicalResource"
    },
    "eventId": "4b9d-b37c-204557d70b97",
    "eventTime": "2024-08-25T12:19:14Z",
    "eventType": "ResourceUsageSpecificationAttributeValueChangeEvent",
    "event": {
        "usageSpecification": {
            "@type": "ResourceUsageSpecification",
            "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/9aa3-3e7737dd5080",
            "id": "9aa3-3e7737dd5080",
            "name": "Voice-Call-Specification",
            "specCharacteristic": [
                {
                    "@type": "CharacteristicSpecification",
                    "name": "originatingCountryCode",
                    "valueType": "string",
                    "description": "country code of the caller",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
```


© TM Forum 2024. All Rights Reserved.

```
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "originatingNumber",
                    "valueType": "string",
                    "description": "phone number of the caller",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "destinationCountryCode",
                    "valueType": "string",
                    "description": "country code of the called party",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "destinationNumber",
                    "valueType": "string",
                    "description": "phone number of the called party",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "duration",
                    "valueType": "string",
                    "description": "duration of the call",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                },
                {
                    "@type": "CharacteristicSpecification",
                    "name": "startDateTime",
                    "valueType": "dateTime",
                    "description": "startdate and starttime of the call",
                    "configurable": true,
                    "minCardinality": 1,
                    "maxCardinality": 1
                }
            ]
        }
    }
}
```


© TM Forum 2024. All Rights Reserved.

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on ResourceUsage

List or find ResourceUsage objects

`GET /resourceUsage?fields=…`

Description

This operation list ResourceUsage entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving previous Usage details for the usageType of 'Voice'.

Request

```
GET
/resourceUsage?fields=status,usageType,usageDate,usageSpecification,resource&usageTy
pe=VOICE
Content-Type: application/json
```

Response

© TM Forum 2024. All Rights Reserved.

```
200

[
    {
        "@type": "ResourceUsage",
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/8cc2840d-b408-4f43-80c0-11da13d84147",
        "id": "8cc2840d-b408-4f43-80c0-11da13d84147",
        "usageDate": "2024-08-26T12:19:14Z",
        "usageType": "DATA",
        "usageCharacteristic": [
            {
                "@type": "StringCharacteristic",
                "name": "direction",
                "valueType": "string",
                "value": "download"
            },
            {
                "@type": "StringCharacteristic",
                "name": "dataTransfered",
                "valueType": "string",
                "value": "45"
            }
        ],
        "relatedParty": [
            {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "role": "Organization",
                "partyOrPartyRole": {
                    "@type": "PartyRef",
                    "href": "http://api.csp.com/tmf-
api/partyManagement/organization/8a41-d6451fe98963",
                    "id": "8a41-d6451fe98963"
                }
            }
        ],
        "resource": {
            "@type": "ResourceRef",
            "href": "https://api.csp.com/tmf-api//resource/qerr-475",
            "id": "qerr-475",
            "name": "Antenna"
        },
        "usageSpecification": {
            "@type": "ResourceUsageSpecificationRef",
            "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
            "id": "af59-b504c742900e",
            "name": "Antenna for 5G slice"
        }
    },
    {
        "@type": "ResourceUsage",
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
        "id": "5130d5131a23",
        "usageDate": "2024-08-25T12:19:14Z",
        "usageType": "DATA",
        "relatedParty": [
            {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "role": "Organization",
                "partyOrPartyRole": {
```

© TM Forum 2024. All Rights Reserved.

```
                    "@type": "PartyRef",
                    "href": "http://api.csp.com/tmf-
api/partyManagement/organization/8a41-d6451fe98963",
                    "id": "8a41-d6451fe98963"
                }
            }
        ],
        "usageCharacteristic": [
            {
                "@type": "StringCharacteristic",
                "name": "direction",
                "valueType": "string",
                "value": "download"
            },
            {
                "@type": "StringCharacteristic",
                "name": "dataTransfered",
                "valueType": "string",
                "value": "20"
            }
        ],
        "resource": {
            "@type": "ResourceRef",
            "href": "https://api.csp.com/tmf-api//resource/qerr-475",
            "id": "qerr-475",
            "name": "Antenna"
        },
        "usageSpecification": {
            "@type": "ResourceUsageSpecificationRef",
            "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
            "id": "af59-b504c742900e",
            "name": "Antenna for 5G slice"
        }
    }
]
```

Retrieves a ResourceUsage by ID

`GET /resourceUsage/{id}?fields=…`

Description

This operation retrieves a ResourceUsage entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

A request for a previous Usage event. The criteria is the Resource Usage id 5130d5131a23

Request

```
GET /resourceUsage/5130d5131a23
Content-Type: application/json
```

© TM Forum 2024. All Rights Reserved.

Response

```
200

{
    "@type": "ResourceUsage",
    "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
    "id": "5130d5131a23",
    "usageDate": "2024-08-25T12:19:14Z",
    "usageType": "DATA",
    "usageCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "direction",
            "valueType": "string",
            "value": "download"
        },
        {
            "@type": "StringCharacteristic",
            "name": "dataTransfered",
            "valueType": "string",
            "value": "20"
        }
    ],
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Organization",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "http://api.csp.com/tmf-
api/partyManagement/organization/8a41-d6451fe98963",
                "id": "8a41-d6451fe98963"
            }
        }
    ],
    "resource": {
        "@type": "ResourceRef",
        "href": "https://api.csp.com/tmf-api//resource/qerr-475",
        "id": "qerr-475",
        "name": "Antenna"
    },
    "usageSpecification": {
        "@type": "ResourceUsageSpecificationRef",
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
        "id": "af59-b504c742900e",
        "name": "Antenna for 5G slice"
    }
}
```

Creates a ResourceUsage

`POST /resourceUsage?fields=…`

Description

This operation creates a ResourceUsage entity.

© TM Forum 2024. All Rights Reserved.

Mandatory Attributes

Mandatory Attributes	Rule
resource	
resource.@type	
resource.id	
usageCharacteristic	
usageCharacteristic.@type	
usageCharacteristic.name	
usageCharacteristic.value	
usageDate	
@type	

Usage samples

Here is an example of a request for creating a specific Usage resource.

Request

```
POST /resourceUsage
Content-Type: application/json

{
    "@type": "ResourceUsage",
    "usageDate": "2024-08-25T12:19:14Z",
    "usageType": "DATA",
    "usageCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "direction",
            "valueType": "string",
            "value": "download"
        },
        {
            "@type": "StringCharacteristic",
            "name": "dataTransfered",
            "valueType": "string",
            "value": "20"
        }
    ],
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Organization",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "http://api.csp.com/tmf-
api/partyManagement/organization/8a41-d6451fe98963",
                "id": "8a41-d6451fe98963"
            }
        }
    ],
```

© TM Forum 2024. All Rights Reserved.

```
    "resource": {
        "@type": "ResourceRef",
        "href": "https://api.csp.com/tmf-api//resource/qerr-475",
        "id": "qerr-475",
        "name": "Antenna"
    },
    "usageSpecification": {
        "@type": "ResourceUsageSpecificationRef",
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
        "id": "af59-b504c742900e",
        "name": "Antenna for 5G slice"
    }
}
```

Response

```
201

{
    "@type": "ResourceUsage",
    "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
    "id": "5130d5131a23",
    "usageDate": "2024-08-25T12:19:14Z",
    "usageType": "DATA",
    "usageCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "direction",
            "valueType": "string",
            "value": "download"
        },
        {
            "@type": "StringCharacteristic",
            "name": "dataTransfered",
            "valueType": "string",
            "value": "20"
        }
    ],
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Organization",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "http://api.csp.com/tmf-
api/partyManagement/organization/8a41-d6451fe98963",
                "id": "8a41-d6451fe98963"
            }
        }
    ],
    "resource": {
        "@type": "ResourceRef",
        "href": "https://api.csp.com/tmf-api//resource/qerr-475",
        "id": "qerr-475",
        "name": "Antenna"
    },
    "usageSpecification": {
        "@type": "ResourceUsageSpecificationRef",
```

© TM Forum 2024. All Rights Reserved.

```
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
        "id": "af59-b504c742900e",
        "name": "Antenna for 5G slice"
    }
}
```

Updates partially a ResourceUsage

PATCH /resourceUsage/{id}?fields=…

Description

This operation allows partial updates of a ResourceUsage entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
usageDate	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
bundledResourceUsage	
description	
externalIdentifier	
isBundle	
relatedParty	
resource	
usageCharacteristic	
usageSpecification	
usageType	

Usage samples

Here is an example of a request for patching a usage record.

© TM Forum 2024. All Rights Reserved.

Request

```
PATCH /resourceUsage/5130d5131a23
Content-Type: application/json

{
    "@type": "ResourceUsage",
    "isBundle": false
}
```

Response

```
200

{
    "@type": "ResourceUsage",
    "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
    "id": "5130d5131a23",
    "isBundle":false,
    "usageDate": "2024-08-25T12:19:14Z",
    "usageType": "DATA",
    "usageCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "direction",
            "valueType": "string",
            "value": "download"
        },
        {
            "@type": "StringCharacteristic",
            "name": "dataTransfered",
            "valueType": "string",
            "value": "22"
        }
    ],
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Organization",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "http://api.csp.com/tmf-
api/partyManagement/organization/8a41-d6451fe98963",
                "id": "8a41-d6451fe98963"
            }
        }
    ],
    "resource": {
        "@type": "ResourceRef",
        "href": "https://api.csp.com/tmf-api//resource/qerr-475",
        "id": "qerr-475",
        "name": "Antenna"
    },
    "usageSpecification": {
        "@type": "ResourceUsageSpecificationRef",
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
        "id": "af59-b504c742900e",
```

© TM Forum 2024. All Rights Reserved.

```
        "name": "Antenna for 5G slice"
    }
}
```

Here is an example of a request for patching a usage record.

Request

```
PATCH /resourceUsage/5130d5131a23
Content-Type: application/merge-patch+json

{
    "@type": "ResourceUsage",
    "isBundle": false
}
```

Response

```
200

{
    "@type": "ResourceUsage",
    "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
    "id": "5130d5131a23",
    "isBundle":false,
    "usageDate": "2024-08-25T12:19:14Z",
    "usageType": "DATA",
    "usageCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "direction",
            "valueType": "string",
            "value": "download"
        },
        {
            "@type": "StringCharacteristic",
            "name": "dataTransfered",
            "valueType": "string",
            "value": "22"
        }
    ],
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Organization",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "http://api.csp.com/tmf-
api/partyManagement/organization/8a41-d6451fe98963",
                "id": "8a41-d6451fe98963"
            }
        }
    ],
    "resource": {
        "@type": "ResourceRef",
        "href": "https://api.csp.com/tmf-api//resource/qerr-475",
```

© TM Forum 2024. All Rights Reserved.

```
        "id": "qerr-475",
        "name": "Antenna"
    },
    "usageSpecification": {
        "@type": "ResourceUsageSpecificationRef",
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
        "id": "af59-b504c742900e",
        "name": "Antenna for 5G slice"
    }
}
```

Here is an example of a request for patching a usage record.

Request

```
PATCH /resourceUsage/5130d5131a23
Content-Type: application/json-patch+json

[
    {
        "op": "replace",
        "path": "/isBundle",
        "value": false
    }
]
```

Response

```
200

{
    "@type": "ResourceUsage",
    "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
    "id": "5130d5131a23",
    "isBundle": false,
    "usageDate": "2024-08-25T12:19:14Z",
    "usageType": "DATA",
    "usageCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "direction",
            "valueType": "string",
            "value": "download"
        },
        {
            "@type": "StringCharacteristic",
            "name": "dataTransfered",
            "valueType": "string",
            "value": "20"
        }
    ],
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Organization",
```

© TM Forum 2024. All Rights Reserved.

```
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "http://api.csp.com/tmf-
api/partyManagement/organization/8a41-d6451fe98963",
                "id": "8a41-d6451fe98963"
            }
        }
    ],
    "resource": {
        "@type": "ResourceRef",
        "href": "https://api.csp.com/tmf-api//resource/qerr-475",
        "id": "qerr-475",
        "name": "Antenna"
    },
    "usageSpecification": {
        "@type": "ResourceUsageSpecificationRef",
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
        "id": "af59-b504c742900e",
        "name": "Antenna for 5G slice"
    }
}
```

Here is an example of a request for patching a usage record.

Request

```
PATCH /resourceUsage/5130d5131a23
Content-Type: application/json-patch-query+json

[
    {
        "op": "replace",
        "path": "/usageCharacteristic/value?name=dataTransfered",
        "value": "22"
    }
]
```

Response

```
200

{
    "@type": "ResourceUsage",
    "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsage/5130d5131a23",
    "id": "5130d5131a23",
    "usageDate": "2024-08-25T12:19:14Z",
    "usageType": "DATA",
    "usageCharacteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "direction",
            "valueType": "string",
            "value": "download"
        },
        {
```

© TM Forum 2024. All Rights Reserved.

```
            "@type": "StringCharacteristic",
            "name": "dataTransfered",
            "valueType": "string",
            "value": "22"
        }
    ],
    "relatedParty": [
        {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "Organization",
            "partyOrPartyRole": {
                "@type": "PartyRef",
                "href": "http://api.csp.com/tmf-
api/partyManagement/organization/8a41-d6451fe98963",
                "id": "8a41-d6451fe98963"
            }
        }
    ],
    "resource": {
        "@type": "ResourceRef",
        "href": "https://api.csp.com/tmf-api//resource/qerr-475",
        "id": "qerr-475",
        "name": "Antenna"
    },
    "usageSpecification": {
        "@type": "ResourceUsageSpecificationRef",
        "href": "https://api.csp.com/tmf-
api/resourceUsageManagement/v5/resourceUsageSpecification/af59-b504c742900e",
        "id": "af59-b504c742900e",
        "name": "Antenna for 5G slice"
    }
}
```


Deletes a ResourceUsage

`DELETE /resourceUsage/{id}`

Description

This operation deletes a ResourceUsage entity.

Operations on ResourceUsageSpecification

Retrieves a ResourceUsageSpecification by ID

`GET /resourceUsageSpecification/{id}?fields=…`

Description

This operation retrieves a ResourceUsageSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

A request for a specific ResourceUsageSpecification with an id 9aa3-3e7737dd5080

© TM Forum 2024. All Rights Reserved.

Request

```
GET /resourceUsageSpecification/9aa3-3e7737dd5080
Content-Type: application/json
```

Response

```
200

{
    "@type": "ResourceUsageSpecification",
    "name": "Antenna for 5G slice",
    "description": "Specification for onsite antenna for 5G slices",
    "version": "2.0",
    "specCharacteristic": [
        {
            "@type": "CharacteristicSpecification",
            "name": "direction",
            "valueType": "string",
            "description": "direction of the data transfer on the antenna",
            "configurable": true,
            "minCardinality": 0,
            "maxCardinality": 2
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "dataTransfered",
            "valueType": "string",
            "description": "volume of the data transfered for the usage.",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        }
    ],
    "lifecycleStatus": "active",
    "validFor": {
        "startDateTime": "2022-02-20T17:00:00Z",
        "endDateTime": "2099-01-11T00:00:00Z"
    }
}
```

List or find ResourceUsageSpecification objects

`GET /resourceUsageSpecification?fields=…`

Description

This operation list ResourceUsageSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieving previous ResourceUsageSpecifications which are at version 2.5

© TM Forum 2024. All Rights Reserved.

Request

```
GET /resourceUsageSpecification?version=2.5
Content-Type: application/json
```

Response

```
200

[
    {
        "@type": "ResourceUsageSpecification",
        "name": "Antenna for 5G slice",
        "description": "Specification for onsite antenna for 5G slices",
        "version": "0.8",
        "specCharacteristic": [
            {
                "@type": "CharacteristicSpecification",
                "name": "direction",
                "valueType": "string",
                "description": "direction of the data transfer on the antenna",
                "configurable": true,
                "minCardinality": 0,
                "maxCardinality": 2
            },
            {
                "@type": "CharacteristicSpecification",
                "name": "dataTransfered",
                "valueType": "string",
                "description": "volume of the data transfered for the usage.",
                "configurable": true,
                "minCardinality": 1,
                "maxCardinality": 1
            }
        ],
        "lifecycleStatus": "inDesign",
        "validFor": {
            "startDateTime": "2022-02-20T17:00:00Z",
            "endDateTime": "2099-01-11T00:00:00Z"
        }
    }
]
```

Creates a ResourceUsageSpecification

`POST /resourceUsageSpecification?fields=…`

Description

This operation creates a ResourceUsageSpecification entity.

Mandatory Attributes

© TM Forum 2024. All Rights Reserved.

Mandatory Attributes	Rule
name	
@type	

Usage samples

Here is an example of a request for creating a specific Resource Usage Specification resource.

Request

```
POST /resourceUsageSpecification
Content-Type: application/json

{
    "@type": "ResourceUsageSpecification",
    "name": "Antenna for 5G slice",
    "description": "Specification for onsite antenna for 5G slices",
    "version": "0.8",
    "specCharacteristic": [
        {
            "@type": "CharacteristicSpecification",
            "name": "direction",
            "valueType": "string",
            "description": "direction of the data transfer on the antenna",
            "configurable": true,
            "minCardinality": 0,
            "maxCardinality": 2
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "dataTransfered",
            "valueType": "string",
            "description": "volume of the data transfered for the usage.",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        }
    ],
    "validFor": {
        "startDateTime": "2022-02-20T17:00:00Z",
        "endDateTime": "2099-01-11T00:00:00Z"
    }
}
```

Response

```
201

{
    "@type": "ResourceUsageSpecification",
    "name": "Antenna for 5G slice",
    "description": "Specification for onsite antenna for 5G slices",
    "version": "0.8",
    "specCharacteristic": [
        {
```

© TM Forum 2024. All Rights Reserved.

```
            "@type": "CharacteristicSpecification",
            "name": "direction",
            "valueType": "string",
            "description": "direction of the data transfer on the antenna",
            "configurable": true,
            "minCardinality": 0,
            "maxCardinality": 2
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "dataTransfered",
            "valueType": "string",
            "description": "volume of the data transfered for the usage.",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        }
    ],
    "lifecycleStatus": "inDesign",
    "validFor": {
        "startDateTime": "2022-02-20T17:00:00Z",
        "endDateTime": "2099-01-11T00:00:00Z"
    }
}
```


Updates partially a ResourceUsageSpecification

`PATCH /resourceUsageSpecification/{id}?fields=…`

Description

This operation allows partial updates of a ResourceUsageSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7396) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	
bundledResourceUsageSpecification	

© TM Forum 2024. All Rights Reserved.

Patchable Attributes	Rule
description	
isBundle	
lastUpdate	
lifecycleStatus	
name	
resourceSpecification	
resourceUsageSpecRelationship	
specCharacteristic	
validFor	
version	

Usage samples

Here is an example of a request for patching a ResourceUsageSpecification record.

Request

```
PATCH /resourceUsageSpecification/9aa3-3e7737dd5080
Content-Type: application/json

{
    "@type": "ResourceUsageSpecification",
    "version": "3.2"
}
```

Response

```
200

{
    "@type": "ResourceUsageSpecification",
    "name": "Antenna for 5G slice",
    "description": "Specification for onsite antenna for 5G slices",
    "version": "3.2",
    "specCharacteristic": [
        {
            "@type": "CharacteristicSpecification",
            "name": "direction",
            "valueType": "string",
            "description": "Direction of the data transfer on the antenna.",
            "configurable": true,
            "minCardinality": 0,
            "maxCardinality": 2
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "dataTransfered",
            "valueType": "string",
            "description": "Volume of the data transfered for the usage.",
```

© TM Forum 2024. All Rights Reserved.

```
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        }
    ],
    "lifecycleStatus": "active",
    "validFor": {
        "startDateTime": "2022-02-20T17:00:00Z",
        "endDateTime": "2099-01-11T00:00:00Z"
    }
}
```

Here is an example of a request for patching a ResourceUsageSpecification record.

Request

```
PATCH /resourceUsageSpecification/9aa3-3e7737dd5080
Content-Type: application/merge-patch+json

{
    "@type": "ResourceUsageSpecification",
    "version": "3.2"
}
```

Response

```
200

{
    "@type": "ResourceUsageSpecification",
    "name": "Antenna for 5G slice",
    "description": "Specification for onsite antenna for 5G slices",
    "version": "3.2",
    "specCharacteristic": [
        {
            "@type": "CharacteristicSpecification",
            "name": "direction",
            "valueType": "string",
            "description": "Direction of the data transfer on the antenna.",
            "configurable": true,
            "minCardinality": 0,
            "maxCardinality": 2
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "dataTransfered",
            "valueType": "string",
            "description": "Volume of the data transfered for the usage.",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        }
    ],
    "lifecycleStatus": "active",
    "validFor": {
        "startDateTime": "2022-02-20T17:00:00Z",
        "endDateTime": "2099-01-11T00:00:00Z"
```

© TM Forum 2024. All Rights Reserved.

```
    }
}
```

Here is an example of a request for patching a ResourceUsageSpecification record.

Request

```
PATCH /resourceUsageSpecification/9aa3-3e7737dd5080
Content-Type: application/json-patch+json

[
    {
        "op": "replace",
        "path": "/version",
        "value": "3.0"
    }
]
```

Response

```
200

{
    "@type": "ResourceUsageSpecification",
    "name": "Antenna for 5G slice",
    "description": "Specification for onsite antenna for 5G slices",
    "version": "3.0",
    "specCharacteristic": [
        {
            "@type": "CharacteristicSpecification",
            "name": "direction",
            "valueType": "string",
            "description": "direction of the data transfer on the antenna",
            "configurable": true,
            "minCardinality": 0,
            "maxCardinality": 2
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "dataTransfered",
            "valueType": "string",
            "description": "volume of the data transfered for the usage.",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        }
    ],
    "lifecycleStatus": "active",
    "validFor": {
        "startDateTime": "2022-02-20T17:00:00Z",
        "endDateTime": "2099-01-11T00:00:00Z"
    }
}
```

Here is an example of a request for patching a ResourceUsageSpecification record.

© TM Forum 2024. All Rights Reserved.

Request

```
PATCH /resourceUsageSpecification/9aa3-3e7737dd5080
Content-Type: application/json-patch-query+json

[
    {
        "op": "replace",
        "path": "/specCharacteristic/description?name=direction",
        "value": "Direction of the data transfer on the antenna."
    }
]
```

Response

```
200

{
    "@type": "ResourceUsageSpecification",
    "name": "Antenna for 5G slice",
    "description": "Specification for onsite antenna for 5G slices",
    "version": "3.0",
    "specCharacteristic": [
        {
            "@type": "CharacteristicSpecification",
            "name": "direction",
            "valueType": "string",
            "description": "Direction of the data transfer on the antenna.",
            "configurable": true,
            "minCardinality": 0,
            "maxCardinality": 2
        },
        {
            "@type": "CharacteristicSpecification",
            "name": "dataTransfered",
            "valueType": "string",
            "description": "Volume of the data transfered for the usage.",
            "configurable": true,
            "minCardinality": 1,
            "maxCardinality": 1
        }
    ],
    "lifecycleStatus": "active",
    "validFor": {
        "startDateTime": "2022-02-20T17:00:00Z",
        "endDateTime": "2099-01-11T00:00:00Z"
    }
}
```

Deletes a ResourceUsageSpecification

`DELETE /resourceUsageSpecification/{id}`

Description

This operation deletes a ResourceUsageSpecification entity.

© TM Forum 2024. All Rights Reserved.

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

© TM Forum 2024. All Rights Reserved.

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

© TM Forum 2024. All Rights Reserved.

```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

© TM Forum 2024. All Rights Reserved.

Acknowledgements

Release History

Release Number	Date	Release led by:	Description
V4.0	July 2021	Kamal Maghsoudlou/Ericsson, Kevin McDonnell/Huawei	1st release of the Usage API
v5.0	June 2024	Sergei Lukin/Deutsche Telekom	migrated to OAS 3.0
v5.0	Dec 2024	Revathi Sivaji/TM Forum, Heidi Lobecker/TM Forum	TM Forum Approved

Contributors to Document

Kamal Maghsoudlou	Ericsson
Kevin McDonnell	Huawei
Pierre Gauthier	TM Forum
Sergei Lukin	Deutsche Telekom
Uwe Sülter	Deutsche Telekom

© TM Forum 2024. All Rights Reserved.